Week 9

Methods

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Check-ins

- Spreadsheet here

What Should Methods Include

- Everything you did
  - Then you start to remove text when you are running out of space
- Excluding content leads to the reproducibility crisis
  - Biomedical Research
  - Software Engineering
  - Social Sciences

Methods Specifics

- Participants
- Materials & Measuers
- Procedures
- Analytic Plan
- More than just this though
  - Additional content depends on your:
    - Field
    - Content area
    - Journal/Guidelines for research paper/poster

Analyses

- Power analyses
  - pwr package in R
  - Power analysis in Python
    - Another link for power analysis in Python
- DO THIS BEFORE ANALYSES
  - Post-hoc power analyses give you a noisy estimate from using observed power (size of effect) vs proposed population effect (hypothetical power)
  - Andrew Gelman's interpretation of post-hoc power analyess

Power Analyses



Figure 1: Photo by Carlos Jamaica

Participants

- Is the same for (quasi) experiments and surveys
- Type of sampling (Random sampling? Random Assignment?)
  - How were people sampled?
  - How many people were sampled?
  - Who are these people?
  - Why were they sampled?
  - When were they sampled?
  - Where were they sampled (generally)?

Participants

- Criteria for participation
- Exclusion criteria
- Cross sectional vs longitudinal
- Is this data part of a larger study?
  - Cite that study

Materials

- This includes how you operationalized variables in your study
- Variables included in the analyses
  - Variable type, how variables are coded
    - Specific coding
  - cross-sectional/longitudinal/cohort-design
- Include how demographic variables were coded
- Measures
  - Need to include reliability and validity of measures
  - Cite references of original measures
- If using measures you created, add more detail about what you used

Procedures

- Everything that you plan to do/are doing/did in your research project
- Write about what you did in your study **in detail**
- Did you introduce some sort of stimuli?
- Was there deception?
- More detail helps with the reporducibility crisis
- How did you separate into groups
- Cross-sectional/longitudinal
  - Pre/post
  - Multiple time points

Analytic Plan

- Data cleaning procedure
- Data processing steps
- Tools/software used
- Model(s) tested
- Fit metrics

Data Cleaning

- How much data was collected originally
  - How much missing data
- Nonresponse rate
- How were outliers dealt with
  - JP Note: Don't remove outliers from data as first step
    - Look into the problem before removing data
- Identifying and removing non-serious responses
  - Figure out if data points are likely to be non-serious responses
- Leave comments on your script/Leave notes on a notebook/document

Data Processing

- Training/test split percentages and method
- Standardizing variables/features
- How was missing data addressed (median, mean, multiple imputation)
- Feature selection/engineering
- Encoding (e.g., dummy/One-hot, etc.)
- Anything you propose here may require citations to support your reasoning

Tools/Software

- What did you use to analyze data?
- Version numbers should be included as well
- Can also provide a link to libraries used in GitHub
- Library citations are also helpful for authors, especially for open-source projects

Model(s)

- Explain why you chose specific models to test
- Make sure the models you are testing are comparable
- Linear regressions without interactions != ensemble models
  - Do extra work in preprocessing to make the models you compare actually comparable

Fit Metrics/Model Evaluation

- Choose the fit metrics BEFORE you run analyses
- Could list as many as you want
- Could list metrics that are consistent with the literature you've read
- Could list metrics consistent with your goal (e.g., precision, recall, f1 score)
- Should have fit metrics that match the bottom two bullet points

Introducing Figures to Methods

- Sometimes your work can only be shown through figures
  - This could be a workflow diagram, a screenshot of stimuli, etc.
- If making a diagram, make sure the font matches your paper's text
- Your figures should be standalone information
  - Captions for clarification should be the only text

DOUBLE CHECK YOUR WORK

- You will forget something GUARANTEED
- Maybe you didn't code a variable correctly
  - Treated a categorical variable as a numeric variable
- Maybe you interpreted a finding on the wrong scale
- Reporting the wrong finding because you were tired
- Accidentally typed out the wrong value, making the finding that much more impressive than it actually is

Let's Review Some Papers

- Break into groups based on type of methods your project will employ
  - AI
  - Predictive analyses
  - "experiment" article

Methods Pebble

- Take 15 minutes to write down what you know about your methods